

TEMPERATE CONTINENTAL STEPPE CLIMATE

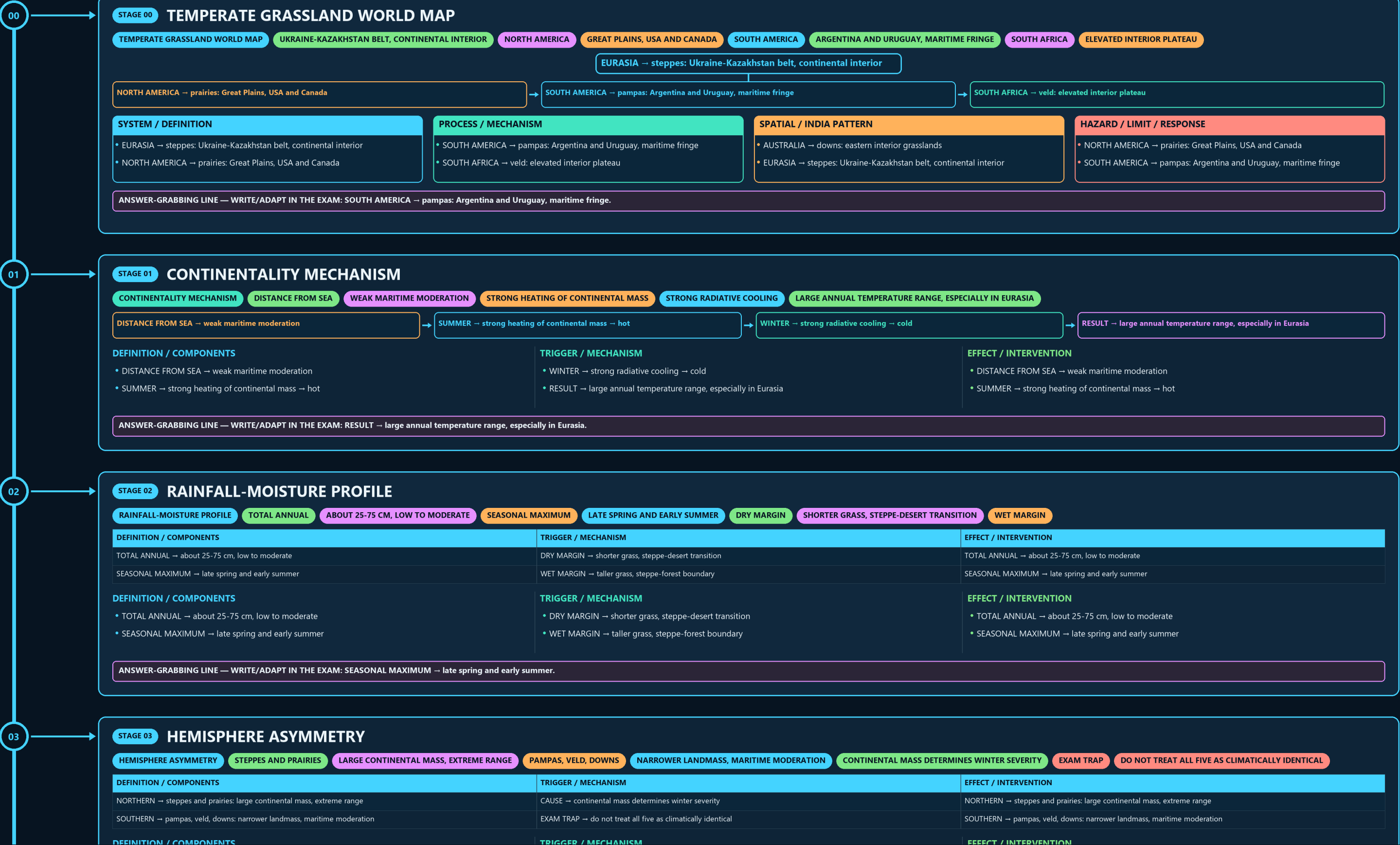
TEMPERATE GRASSLAND WORLD MAP → CONTINENTALITY MECHANISM → RAINFALL-MOISTURE PROFILE → HEMISPHERE ASYMMETRY → CHERNOZEM-SOIL CHAIN → GRANARY CONVERSION TIMELINE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

SEASONAL MAXIMUM → late spring and early summer

WET MARGIN → taller grass, steppe-forest boundary

SEASONAL MAXIMUM → late spring and early summer

DEFINITION / COMPONENTS

TOTAL ANNUAL → about 25-75 cm, low to moderate

SEASONAL MAXIMUM → late spring and early summer

TRIGGER / MECHANISM

DRY MARGIN → shorter grass, steppe-desert transition

WET MARGIN → taller grass, steppe-forest boundary

EFFECT / INTERVENTION

TOTAL ANNUAL → about 25-75 cm, low to moderate

SEASONAL MAXIMUM → late spring and early summer

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: SEASONAL MAXIMUM → late spring and early summer.

03

STAGE 03

HEMISPHERE ASYMMETRY

HEMISPHERE ASYMMETRY

STEPPES AND PRAIRIES

LARGE CONTINENTAL MASS, EXTREME RANGE

PAMPAS, VELD, DOWNS

NARROWER LANDMASS, MARITIME MODERATION

CONTINENTAL MASS DETERMINES WINTER SEVERITY

EXAM TRAP

DO NOT TREAT ALL FIVE AS CLIMATICALLY IDENTICAL

DEFINITION / COMPONENTS

NORTHERN → steppes and prairies: large continental mass, extreme range

SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

TRIGGER / MECHANISM

CAUSE → continental mass determines winter severity

EXAM TRAP → do not treat all five as climatically identical

EFFECT / INTERVENTION

NORTHERN → steppes and prairies: large continental mass, extreme range

SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

DEFINITION / COMPONENTS

NORTHERN → steppes and prairies: large continental mass, extreme range

SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

TRIGGER / MECHANISM

CAUSE → continental mass determines winter severity

EXAM TRAP → do not treat all five as climatically identical

EFFECT / INTERVENTION

NORTHERN → steppes and prairies: large continental mass, extreme range

SOUTHERN → pampas, veld, downs: narrower landmass, maritime moderation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXAM TRAP → do not treat all five as climatically identical.

04

STAGE 04

CHERNOZEM-SOIL CHAIN

CHERNOZEM-SOIL CHAIN

GRASSLAND COVER

DEEP ROOT NETWORK DECAYS IN PLACE

LOW LEACHING

BASE-RICH CONDITIONS PRESERVED

HUMUS ACCUMULATES

DARK, DEEP, FERTILE SOIL PROFILE

CHERNOZEM OR BLACK EARTH, AMONG WORLD'S MOST FERTILE

GRASSLAND COVER → deep root network decays in place

LOW LEACHING → base-rich conditions preserved

HUMUS ACCUMULATES → dark, deep, fertile soil profile

RESULT → chernozem or black earth, among world's most fertile

DEFINITION / COMPONENTS

GRASSLAND COVER → deep root network decays in place

LOW LEACHING → base-rich conditions preserved

TRIGGER / MECHANISM

HUMUS ACCUMULATES → dark, deep, fertile soil profile

RESULT → chernozem or black earth, among world's most fertile

EFFECT / INTERVENTION

GRASSLAND COVER → deep root network decays in place

LOW LEACHING → base-rich conditions preserved

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → chernozem or black earth, among world's most fertile.

05

STAGE 05

GRANARY CONVERSION TIMELINE

GRANARY CONVERSION TIMELINE

NATURAL STATE

PASTORAL GRASSLAND, LOW-DENSITY USE

RAIL + OCEAN FREIGHT

ACCESS TO DISTANT INDUSTRIAL MARKETS

MECHANISATION + HYV

HIGH OUTPUT PER WORKER ON LEVEL PLAINS

WORLD'S WHEAT GRANARIES

NATURAL STATE → pastoral grassland, low-density use

RAIL + OCEAN FREIGHT → access to distant industrial markets

MECHANISATION + HYV → high output per worker on level plains

RESULT → world's wheat granaries: Ukraine, prairies, pampas

DEFINITION / COMPONENTS

NATURAL STATE → pastoral grassland, low-density use

RAIL + OCEAN FREIGHT → access to distant industrial markets

TRIGGER / MECHANISM

MECHANISATION + HYV → high output per worker on level plains

RESULT → world's wheat granaries: Ukraine, prairies, pampas

EFFECT / INTERVENTION

NATURAL STATE → pastoral grassland, low-density use

RAIL + OCEAN FREIGHT → access to distant industrial markets

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESULT → world's wheat granaries: Ukraine, prairies, pampas.

06

STAGE 06

ANTI-DETERMINISM FIREWALL

ANTI-DETERMINISM FIREWALL

PHYSICAL BASE

FERTILE SOIL + LEVEL RELIEF + SUITABLE CLIMATE

CONVERSION AGENTS

RAIL, FREIGHT, MECHANISATION, MARKET DEMAND

WITHOUT ACCESS

SAME SOIL, NO GRANARY FOR CENTURIES

ENDOWMENT IS NECESSARY, ACCESS IS SUFFICIENT

PHYSICAL BASE → fertile soil + level relief + suitable climate

CONVERSION AGENTS → rail, freight, mechanisation, market demand

WITHOUT ACCESS → same soil, no granary for centuries

CONCLUSION → endowment is necessary, access is sufficient

DEFINITION / COMPONENTS

PHYSICAL BASE → fertile soil + level relief + suitable climate

CONVERSION AGENTS → rail, freight, mechanisation, market demand

TRIGGER / MECHANISM

WITHOUT ACCESS → same soil, no granary for centuries

CONCLUSION → endowment is necessary, access is sufficient

EFFECT / INTERVENTION

PHYSICAL BASE → fertile soil + level relief + suitable climate

CONVERSION AGENTS → rail, freight, mechanisation, market demand

TEMPERATE CONTINENTAL STEPPE CLIMATE • TILE 2/4 • same master rows 2363-5597 of 10324 • continues below

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ENABLERS → HYV seeds + irrigation + fertiliser + MSP procurement.

09

STAGE 09 GREEN REVOLUTION PACKAGE

GREEN REVOLUTION PACKAGE HYV SEMI-DWARF WHEAT RAISED YIELD POTENTIAL CANAL + TUBE-WELL IRRIGATION REDUCED RAINFALL DEPENDENCE CHEMICAL FERTILISERS SUPPORTED INTENSIVE CULTIVATION ASSURED FLOOR PRICE AND CENTRAL POOL STOCKS

HYV SEMI-DWARF WHEAT → raised yield potential

CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

CHEMICAL FERTILISERS → supported intensive cultivation

MSP/PROCUREMENT → assured floor price and central pool stocks

DEFINITION / COMPONENTS

- HYV SEMI-DWARF WHEAT → raised yield potential
- CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

TRIGGER / MECHANISM

- CHEMICAL FERTILISERS → supported intensive cultivation
- MSP/PROCUREMENT → assured floor price and central pool stocks

EFFECT / INTERVENTION

- HYV SEMI-DWARF WHEAT → raised yield potential
- CANAL + TUBE-WELL IRRIGATION → reduced rainfall dependence

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MSP/PROCUREMENT → assured floor price and central pool stocks.

10

STAGE 10 SAVANNA-STEPPE COMPARISON

SAVANNA-STEPPE COMPARISON TROPICAL SAVANNA VERSUS TEMPERATE STEPPE/PRAIRIE WET-AND-DRY SEASONAL VERSUS LOW-TOTAL SPRING MAXIMUM LEACHED LATERISING VERSUS BASE-RICH CHERNOZEM/PRAIRIE PASTORAL-SUBSISTENCE VERSUS MECHANISED-COMMERCIAL GRAIN

DEFINITION / COMPONENTS

LATITUDE → tropical savanna versus temperate steppe/prairie
RAINFALL → wet-and-dry seasonal versus low-total spring maximum

TRIGGER / MECHANISM

SOIL → leached laterising versus base-rich chernozem/prairie
ECONOMY → pastoral-subsistence versus mechanised-commercial grain

EFFECT / INTERVENTION

LATITUDE → tropical savanna versus temperate steppe/prairie
RAINFALL → wet-and-dry seasonal versus low-total spring maximum

DEFINITION / COMPONENTS

- LATITUDE → tropical savanna versus temperate steppe/prairie
- RAINFALL → wet-and-dry seasonal versus low-total spring maximum

TRIGGER / MECHANISM

- SOIL → leached laterising versus base-rich chernozem/prairie
- ECONOMY → pastoral-subsistence versus mechanised-commercial grain

EFFECT / INTERVENTION

- LATITUDE → tropical savanna versus temperate steppe/prairie
- RAINFALL → wet-and-dry seasonal versus low-total spring maximum

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RAINFALL → wet-and-dry seasonal versus low-total spring maximum.

11

STAGE 11 STEPPE ANSWER SPINE

STEPPE ANSWER SPINE CONTINENTAL INTERIOR GRASSLAND WITH LARGE ANNUAL RANGE FIVE REGIONAL NAMES ON A WORLD MAP MOISTURE DEFICIT, CHERNOZEM, GRANARY CONVERSION ACCESS CONVERTED ENDOWMENT INTO RESOURCE

DEFINE → continental interior grassland with large annual range

LOCATE → five regional names on a world map

EXPLAIN → moisture deficit, chernozem, granary conversion

QUALIFY → anti-determinism: access converted endowment into resource

DEFINITION / COMPONENTS

- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

TRIGGER / MECHANISM

- EXPLAIN → moisture deficit, chernozem, granary conversion
- QUALIFY → anti-determinism: access converted endowment into resource

EFFECT / INTERVENTION

- DEFINE → continental interior grassland with large annual range
- LOCATE → five regional names on a world map

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → anti-determinism: access converted endowment into resource.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT INDIA'S WHEAT GRANARY PUNJAB, HARYANA, WESTERN UP INDO-GANGETIC PLAIN. PUNJAB IS DESCRIBED AS A MAJOR FOOD BASKET/GRANARY IN PUNJAB'S HIGH CULTIVATED SHARE AND CROPPING INTENSITY ARE SOURCE-ERA
HARYANA'S WHEAT BELT SPANS IRRIGATED EASTERN, CENTRAL AND WESTERN WHEAT IS A RABI CROP

OPTIONAL PROCESS DEPTH

- India's wheat granary = Punjab, Haryana, western UP / Indo-Gangetic plain.
- Punjab is described as a major food basket/granary in the source text.
- Punjab's high cultivated share and cropping intensity are source-era indicators, not current statistics.
- Haryana's wheat belt spans irrigated eastern, central and western plains; use current district maps.

DATA / INSTITUTION LIMIT

- Wheat is a Rabi crop: sown Oct–Dec, harvested Mar–Apr in source notes.
- Green Revolution package: HYV + irrigation + fertiliser + MSP/procurement.
- Soil is alluvial , not steppe chernozem.
- The source-era text records a substantial marketed surplus; current share requires a dated source.

CONTEMPORARY PARALLEL

- Surplus states: Punjab, Haryana, UP, Rajasthan, MP.
- Deficit destinations include Maharashtra, West Bengal, Bihar and Delhi.
- NOT: India's wheat belt sits on chernozem like the steppe → It sits on Indo-Gangetic alluvial soil.
- NOT: Wheat is a Kharif monsoon crop → Wheat is a Rabi winter crop.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.